

Which Damage Do the Images Support? Evidence-Bounded 3D Damage Estimation on Turbine Blades

Anonymous CVPR submission

Paper ID *****

Abstract

Visual inspection of aero-engine blades must decide whether a change seen in an image is physical damage, where it is on the part, and how deep it goes. Existing methods flag anomalies in 2D, output boxes or voxels, or require a measured 3D scan, and none states whether the available views are sufficient to support a claimed shape change. We study evidence-bounded 3D damage estimation: given a healthy CAD model and a few calibrated images, output a displacement field on the blade surface when the images support one, and abstain when camera error, lighting or a surface-colour change explain the images equally well. Our estimator compares a nuisance-only explanation (per-view pose, light gains, an albedo patch) with the same explanation plus a continuous dent, using a differentiable renderer with a neural shader. Dent hypotheses are proposed by a Rao score test computed for every surface location in one backward pass, and the decision threshold on the resulting fit gain is calibrated on healthy and colour-only blades with a split-conformal guarantee. We release BladeDelta, a benchmark of [TODO: N] rendered inspection cases on [TODO: N] public Rotor37 blade geometries with exact damaged meshes, colour-only negatives, three dent families and controlled capture perturbations, with a sealed test split. [TODO: one sentence with headline test numbers]

1. Introduction

Turbine and compressor blades are inspected with cameras, and the inspector's question is physical: is there a dent, where on the blade is it, and how deep is it? Damage of interest is small. On a blade imaged at 512×512 pixels, a dent of 0.07 mm changes a few hundred pixels by two to seven grey levels. A one-degree camera error, a ten-percent lighting change or a stain changes thousands.

Current vision methods answer an easier question. Industrial anomaly detection localises unusual pixels [1, 18], including under pose and illumination change [3, 11]. Reference-based methods compare a photograph with a CAD model and output boxes [2], or lift per-view scores into an anomaly volume [7]. CAD-conditioned reconstruction fits a deformation field to sparse posed views [17, 21], and leaves measurement-quality damaged surfaces to future work [17]. None of these outputs states whether the images actually support a physical change, as opposed to a nuisance explanation.

Our own earlier experiments made this concrete. A render-and-compare fit on public Rotor37 blade geometry produced false dents on 14 of 14 healthy or colour-only blades under a one-degree camera error, and on unfamiliar dents the best image match selected a worse 3D shape in 8 of 12 depth sweeps. Better image agreement is not better geometry, and a plausible reconstruction can choose the wrong physical explanation.

We therefore formulate evidence-bounded 3D damage estimation. The input is a healthy CAD mesh and K calibrated images; the output is a per-vertex displacement field together with a decision, supported or not supported by these views. Our estimator treats this as a nested hypothesis comparison. H_0 explains the images with nuisance only: per-view pose corrections, light gains and a surface albedo patch. H_1 adds a continuous dent. The fit gain $L(H_0) - L(H_1)$ is thresholded at a value calibrated on negatives, giving a controlled false-change rate.

Making this work required separating the roles of the evidence. Pose is fitted from silhouettes, which do not depend on lighting; the dent is compared against nuisance only on interior surface pixels that are not grazing, not near a silhouette and not on the blade root. Dent hypotheses come from a Rao score test evaluated for every surface location and radius with one backward pass through the renderer, and the dent and colour-patch explanations are searched symmetri-

076	cally, so a colour patch fitted first cannot hide a dent.	
077	We also report observability: on our development split,	
078	70% of dents placed on camera-facing surfaces are less	
079	than half visible in the two input views, and the correct	
080	output for those is to abstain.	
081	Contributions:	
082	• The task of evidence-bounded 3D damage estimation	125
083	from sparse views and healthy CAD, with metrics	126
084	that separate localisation, shape accuracy, false	127
085	physical change and abstention on unobservable	128
086	damage.	129
087	• An estimator combining a differentiable renderer	130
088	with a neural shader, silhouette-only pose, interior-	131
089	domain nested hypothesis testing, score-test propos-	
090	als and split-conformal calibration.	
091	• BladeDelta, a benchmark built on public Rotor37	
092	geometry [4] with exact damaged meshes, colour-	
093	only negatives, three dent families that differ from	
094	the estimator's own profile, and controlled camera	
095	and lighting perturbations, with a protocol and	
096	sealed test split fixed before evaluation.	
097	2. Related Work	
098	Industrial and 3D anomaly detection. Memory-bank	
099	methods such as PatchCore [18] localise anomalous pixels;	
100	MVTec 3D-AD [1] and Anomaly-ShapeNet [13]	
101	add measured or synthetic 3D. Pose- and illumination-	
102	agnostic settings use Gaussian-splat references [11]	
103	[TODO: cite PIAD after verifying the primary source],	
104	and M ² AD [3] benchmarks view-illumination inter-	
105	play. Multi-view settings include RAD [5] and	
106	Sim3D [7], which lifts per-view scores into an anomaly	
107	volume; recent work warns that adding views can leak	
108	information in multi-view evaluation [6]. Synthetic	
109	anomalies have been argued sufficient for 3D detec-	
110	tion [20], and incomplete-modality methods train with	
111	3D and test on RGB [19]. These methods output	
112	anomaly scores; they do not estimate a physical sur-	
113	face change or state whether the views support one.	
114	Reference CAD and blade inspection. Looking 3D [2]	
115	aligns a query photograph with a reference shape and	
116	localises abnormal regions with boxes. BladeSynth [8]	
117	renders procedural nicks, dents, scratches and corrosion	
118	on aero-engine blades for 2D detection. Model-free uni-	
119	versal detection reconstructs a normal shape by GAN	
120	inversion [15]. We instead output a 3D displacement	
121	field on a known design.	
122	CAD-conditioned reconstruction. CAD-NeRF [21] re-	
123	trieves a CAD model and jointly optimises deformation,	
124	pose and appearance from few uncalibrated views.	
	CADsplat [17] fits a shared deformation field to sparse	125
	posed views with a CAD prior and explicitly leaves	126
	measurement-quality deformed surfaces to future work.	127
	Our free-form deformation baseline follows this family.	128
	The difference is the output: a decision on whether	129
	the deformation is supported, with a calibrated false-	130
	change rate.	131
	Inverse rendering and uncertainty. Inverse rendering	132
	separates shape and material from images [16, 22];	133
	we use differentiable rasterisation [12] with a learned	134
	shader for the same purpose on a known geometry.	135
	Radiance-field uncertainty [9, 10] and observation com-	136
	pleteness [14] quantify where a reconstruction is unre-	137
	liable. They do not test a physical-change hypothesis	138
	against nuisance explanations, which is the question an	139
	inspector needs answered.	140
	3. Method	141
	Setup. We are given a healthy mesh $\mathcal{M} = (V, F)$ with	142
	vertices $V \in \mathbb{R}^{N \times 3}$, K observed images $I_1, \dots, I_K$ with	143
	nominal intrinsics and extrinsics, and nominal light-	144
	ing. We seek a displacement field $D \in \mathbb{R}^{N \times 3}$ on the	145
	airfoil and a decision $s \in \{\text{supported}, \text{not supported}\}$.	146
	Nuisance variables η are a per-view pose correction (a	147
	rotation about the part and a camera-frame transla-	148
	tion), three light gains, and an optional albedo patch.	149
	Differentiable renderer. We rasterise $\mathcal{M} + D$ with nvd-	150
	iffrast [12] using smooth vertex normals. A neural	151
	shader maps the per-pixel world normal n , position x	152
	and view direction ω to one radiance component per	153
	light, $R_k(n, x, \omega)$, and an environment term R_0 . Be-	154
	cause light transport is linear in each light's intensity,	155
	the pixel colour is $T(a \odot (R_0 + \sum_k g_k R_k))$ for albedo	156
	multiplier a , light gains g and a learned monotone tone	157
	curve T , followed by anti-aliasing and the sensor's pixel	158
	filter. An unstructured shader that took g as an input	159
	learned to ignore it, and lighting changes then pro-	160
	duced spurious dents. The shader is trained once, on	161
	renders of blade geometries disjoint from all evaluation	162
	geometries, with a paired loss on image differences be-	163
	tween lighting settings. The healthy CAD rendered un-	164
	der the nominal capture provides a per-pixel correction	165
	$B = I^{\text{ref}} - \hat{I}(\mathcal{M}; \eta_0)$ that absorbs static shader error.	166
	On a held-out blade its mean error is 0.92 grey levels;	167
	image changes caused by a lighting change correlate	168
	with the reference renderer at 0.90–0.98, and changes	169
	caused by a dent at 0.96–0.98 in the 120° view.	170
	Nuisance fit (H_0). Each view's pose is initialised by	171
	Euclidean ECC alignment of the observed and refer-	172

ence images, then refined by matching anti-aliased silhouettes to the observed foreground, coarse to fine. Silhouettes do not depend on lighting, which prevents the pose from compensating for a brightness change. With pose fixed, light gains are fitted on the full image.

Evidence domain. All further comparisons use a fixed interior domain Ω : airfoil pixels at least 6 px from any silhouette or self-occlusion edge, excluding the blade root and pixels where $|n \cdot \omega| < 0.35$. Both hypotheses are evaluated on the same Ω , so their losses are comparable.

Dent model (H_1). A dent is parameterised by a sub-vertex centre c (an anchor vertex plus a tangent offset), depth $\delta \geq 0$, radius r , aspect ratio, orientation, profile exponent p and a signed rim coefficient κ :

$$D_i = -\delta \left[(1 - s_i^2)_+^p - \frac{\kappa}{2} \rho(s_i) \right] n_c, \quad (1)$$

where s_i is the anisotropic normalised distance of vertex i from c in the tangent plane, ρ a ring profile and n_c the centre normal, restricted to same-side airfoil vertices. None of the benchmark’s truth families matches this profile exactly.

Score-test proposals. Let $h \in \mathbb{R}^N$ be an inward normal displacement. One backward pass gives $g = -\partial L_\Omega / \partial h$ at $h = 0$. For a kernel $w_{c,r}$ centred at c with radius r , the normalised score $\langle w_{c,r}, g \rangle / \|w_{c,r}\|$ is the Rao score statistic for a small dent at (c, r) against H_0 . We evaluate it for all candidate centres and two radii with a sparse matrix product, and start H_1 fits from the top- k after non-maximum suppression. Proposals are computed from the nuisance-only residual, and each start is fitted both with and without the H_0 albedo patch, so a colour explanation fitted first cannot hide a dent.

Decision. The fit gain is $\Delta = L_\Omega(H_0) - L_\Omega(H_1) \geq 0$, as H_1 nests H_0 . We report D if $\Delta > \tau$ and $D = 0$ otherwise, with τ the split-conformal $(1-\alpha)$ quantile of Δ on calibration negatives (healthy and colour-only blades), which bounds the false-change rate on exchangeable negatives by α . We use a Charbonnier loss ($\epsilon = 2/255$) summed over Ω ; a Student- t noise model and a wider Charbonnier floor did not improve separation on development data.

4. The BladeDelta Benchmark

Geometry. Healthy blades are public Rotor37 compressor geometries [4] (CC BY-SA 4.0), normalised to

Split	Blades	Healthy	Stain	Dent	Dent + stain
Shader training	13	only for the renderer; never evaluated			
Development	12	12	24	36	24
Calibration	12	24	36	36	24
Test (sealed)	30	30	60	90	60

Table 1. BladeDelta case counts. Each case has 7 rendered views; methods use 2 by default.

unit span and closed with a labelled artificial hub collar. Each mesh has 29,989 vertices. Shapes 0–12 were used during earlier development and only to train the shader. Shapes 13–24 form the development split, 25–36 the calibration split and 37–66 the sealed test split; 67–109 are held in reserve.

Damage. Each blade receives one of: no change, a colour-only patch (a soft darkened or rust-tinted albedo change), a dent, or a dent plus a colour patch (half overlapping). Dents follow one of three families — a cubic profile with a raised rim, a Gaussian, and an anisotropic elliptic profile (aspect 0.4–1) — with depth log-uniform in $[0.0005, 0.006]$ blade spans (about 0.04–0.45 mm on a 74 mm Rotor37 blade) and radius in $[0.04, 0.12]$. Centres face at least one of seven cameras and lie at least $\max(0.1, 1.2r)$ from the airfoil boundary.

Capture. Images are rendered with Blender Cycles (32 samples, denoised, AgX tone mapping, 512×512) from seven cameras on a ring around the blade, under one of four conditions: nominal; camera jitter (yaw, pitch and roll each uniform in $\pm 1.5^\circ$ per view); light jitter (each of three area lights $\times [0.85, 1.15]$); or both. Methods receive only the nominal cameras and lighting. The default input is two views (60° and 120°); four- and seven-view settings are ablations.

Protocol. The split, case lists, metrics and hypotheses were hash-frozen before any test image was rendered; protocol changes are logged with dates. For each damaged case we record observability, the depth-weighted fraction of the dent visible in the input views.

CAD6 transfer split. To test transfer beyond Rotor37 we add six industrial blade designs supplied as STEP solids (airfoil spans 24–70 mm), with platforms, fillets and dovetail roots. The airfoil is the largest B-rep face (two faces on one blade); everything else is excluded from evidence like the Rotor37 collar. Each blade is normalised to unit airfoil span in the Rotor37 frame

and receives 76 cases (456 in total): 30% dents centred where both input cameras see the surface, 15% dents hidden from both, 20% healthy, 15% stain only, 10% dent plus stain and 10% healthy with camera error alone ($\pm 0.5^\circ$, $\pm 1^\circ$, $\pm 1.5^\circ$). Dents are sized in millimetres as an inspector would specify them: depth log-uniform in $[0.1, 1.0]$ mm and radius in $[1, 5]$ mm. The case list was hashed before rendering, and no component — shader, threshold or hyperparameter — was refit on this split.

Metrics. Relative local error is the RMS vertex error over the true dent support divided by that of the unchanged CAD (1: no better than CAD; 0: exact). We also report relative whole-airfoil error, depth error, centre error, detection, false physical change on negatives (any vertex moved more than 2×10^{-5}) and abstention on unobservable dents, with 95% bootstrap intervals resampled by blade shape.

5. Experiments

Baselines. Unchanged CAD reports the healthy mesh. Discrete search re-implements our earlier method: image-guided proposals, a fixed cubic profile of radius 0.08 and depths $\{0.0005, 0.001, 0.0015\}$, at nominal pose. Free-form field fits a per-vertex normal displacement with Laplacian smoothness and ℓ_1 sparsity after the same pose and lighting fit. MLP deformation field replaces the per-vertex values with a coordinate MLP (positional encoding, two hidden layers), the continuous deformation field used by CAD-prior reconstruction. These two baselines reimplement the core idea of CADsplat [17] — a smooth deformation field fit to sparse posed views under a CAD prior — within our renderer; they are not the official model, whose code had not been released at the time of writing. 2D detector, lifted is a U-Net trained on (photo, healthy render) pairs of separate exposed blades to segment dents; detections are back-projected onto the mesh through the nominal cameras. All methods share the renderer, cameras and observations, and each is thresholded by the same calibration procedure on its own score.

Main results.

Risk-coverage.

Ablations.

Why recall is limited. Recall is set by dent depth and by the false-alarm budget, not by a failure to locate (Table 5). Every reported dent is in the right place,

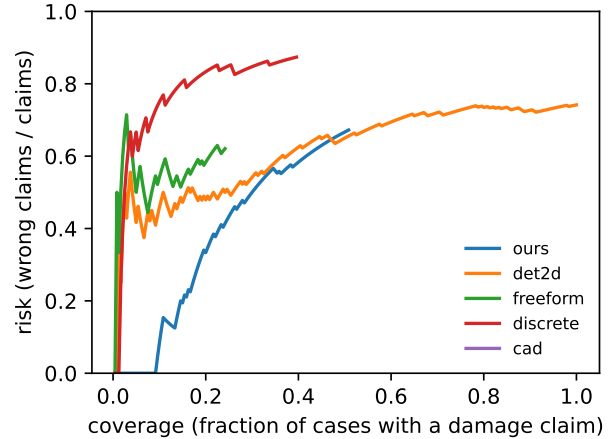


Figure 1. Risk (share of damage claims that are wrong) against coverage (share of cases with a claim) as the threshold varies, sealed test.

and 12/13 visible dents deeper than 0.2mm are reported; most misses are shallower than 0.1 mm, where the change is about one grey level. In 6 of the 17 misses the dent hypothesis converged on the true location, but its evidence stayed below $\tau = 1.52$. The threshold is held up by appearance changes: stains account for 8 of the 9 false alarms, with stain-only fit gains up to 4.7 (5.6 for one healthy blade under camera error). Lowering τ to 0.54 would reach 70% recall at a 20% false-alarm rate. The misses are therefore the price of the false-alarm guarantee: an inspector receives few spurious damage claims, and dents below about 0.1 mm are reported as unsupported rather than guessed.

Transfer to unseen blade designs. Table 6 applies every method, unchanged, to the six CAD6 blades (456 cases). Separation falls from 0.82 on the Rotor37 sealed test to 0.74 (transfer gap 0.08), with 94% of visible dents found, 67% false alarms and 33% abstention on hidden dents. The threshold was calibrated on Rotor37 only, so the false-alarm rate here measures how far the guarantee survives a change of geometry, not a property the method was fit to. Per blade (Table 7), dents are still ranked above negatives (best AUROC on blade 1 (24 mm, AUROC 0.98), worst on blade 3 (31 mm, AUROC 0.64)), but the frozen threshold fails on the smaller blades: false alarms fall with airfoil span (Spearman $\rho = -0.88$, $p = 0.02$). Across blades, airfoil span and detection rate have Spearman $\rho = -0.93$ ($p = 0.01$; six blades, so weak evidence): 99% of visible dents were found on the three smaller blades and 88% on the three larger ones. Because dents are specified in millimetres, the same dent is relatively deeper

Method	Rel. local err. (obs.) ↓	Detection (obs.) % ↑	False change % ↓	Abstain (unobs.) % ↑	Centre err. ↓
BladeDelta (ours)	0.69 [0.59, 0.79]	54 [39, 68]	10 [4, 17]	85 [77, 91]	0.01 [0.00, 0.02]
Free-form field	1.00 [1.00, 1.00]	14 [4, 24]	4 [1, 9]	99 [97, 100]	0.14 [0.04, 0.34]
Discrete search	1.00 [0.99, 1.00]	5 [0, 13]	6 [1, 10]	98 [95, 100]	0.02 [0.02, 0.02]
Unchanged CAD	1.00 [1.00, 1.00]	0 [0, 0]	0 [0, 0]	100 [100, 100]	—

Table 2. Sealed test split. Mean [95% shape-bootstrap CI]. Observable: $\geq 50\%$ of the dent visible in the two input views. Thresholds calibrated on the calibration split (split-conformal, $\alpha = 0.05$).

Views	Observable dents	AUROC ↑	Located % ↑	Method	Rel. local err. ↓	AUROC	Found (%)	Located % ↑
2 views	10/60	0.847	60	Ours, Rotor37 sealed test	0.82	0.82 [0.71, 0.90]	54 [39, 68]	54 [39, 68]
4 views	17/60	0.777	41	Ours, 2D detector, lifted	0.74	0.74 [0.67, 0.84]	94 [89, 98]	70 [65, 75]
7 views	29/60	0.857	62	MLP deformation field	0.86	0.86 [0.83, 0.89]	42 [35, 49]	36 [31, 43]
				Free-form field	0.73	0.73 [0.68, 0.81]	84 [70, 97]	54 [49, 59]
				Discrete search	0.60	0.60 [0.56, 0.66]	61 [36, 84]	37 [32, 42]
				Unchanged CAD	0.65	0.65 [0.58, 0.72]	16 [12, 18]	13 [10, 16]
				Ours v1 (frozen), held-out blades 3–6	0.50	0.50 [0.50, 0.50]	0 [0, 0]	0 [0, 0]

Table 3. More views make more dents observable and the evidence clearer.

Variant	AUROC ↑	Located % ↑	Method	Rel. local err. ↓	AUROC	Found (%)	Located % ↑
Full method	0.855	72	Ours v1.1, held-out blades 3–6	0.72	0.72 [0.65, 0.90]	90 [85, 95]	67 [55, 79]
No camera (pose) correction	0.679	72	Ours, CAD6 leave-one-blade-out τ	0.95	0.95 [0.93, 0.98]	86 [77, 95]	83 [71, 95]
No stain (colour) model	0.753	83		0.70	0.70 [0.63, 0.84]	13 [3, 26]	6 [1, 13]
Whole image instead of interior evidence	0.685	61					
No score-test proposals	0.420	72					
No low-frequency brightness field	0.846	72					

Table 4. Ablations on the development subset (18 observable dents, 18 healthy or stained blades).

Table 6. Transfer to six unseen real blade designs (CAD6, 656 cases; dents 0.1–1.0 mm deep). Every method is frozen: shader trained on Rotor37 only; thresholds from the Rotor37 calibration split. Top row: ours on the Rotor37 sealed test, for the transfer gap. Brackets: 95% intervals resampling blades.

True depth (mm)	2 views	2 views (same cases)	Blades	Airfoil span (mm)	Cases	Visible dents	AUROC	Found (%)
< 0.1	1/10 (1 loc.)	0/4 (0 loc.)	1/5 (0 loc.)	3/21 (3 loc.)	76	31	0.98	100
0.1–0.2	7/14 (7 loc.)	1/2 (1 loc.)	2/5 (2 loc.)	7/26 (7 loc.)	76	31	0.72	100
> 0.2	12/13 (12 loc.)	4/4 (4 loc.)	4/5 (4 loc.)	8/31 (6 loc.)	76	31	0.64	97
All visible dents	20/37 (20 loc.)	5/10 (5 loc.)	7/17 (6 loc.)	18/49 (16 loc.)	76	31	0.66	94
			5	52	76	31	0.93	84
			6	70	76	31	0.86	87

Table 5. Visible test dents reported (located within 0.1 span) by true depth, at the frozen threshold. Below 0.1 mm a dent changes the image by about one grey level; the method declines rather than guess.

Table 7. Ours on each CAD6 blade, smallest to largest airfoil (frozen threshold).

on a small blade; detection tracks depth relative to the airfoil span ($\rho = 0.10$) more strongly than depth in millimetres ($\rho = 0.02$), rising from 92% in the shallowest relative-depth tertile to 97% in the deepest. Diagnosis and fix (v1.1). On a healthy CAD6 capture whose image is identical to the reference, the silhouette pose fit moves a correct camera by about 0.3° and a spurious dent absorbs the misregistration; under real camera error, the image-space reference bias — large on unseen geometry because the shader was trained on Rotor37 only — stays fixed while the blade moves. v1.1 corrects both: it applies the reference correction to the silhou-

ette as well as to shading, and it reprojects the shading correction through the surface so that it moves with the blade (identical to v1 at the nominal pose). Both changes were developed on CAD6 blades 1–2 only and frozen, with a threshold recalibrated on the Rotor37 calibration split, before blades 3–6 were run. On those held-out blades v1.1 raises AUROC from 0.72 to 0.95, cuts false alarms from 51% to 4%, locates 83% of visible dents (v1: 67%) and abstains on 98% of hidden dents (v1: 50%). Recalibrating the threshold on CAD6 negatives, leave-one-blade-out, gives 13% found at 4% false alarms: the cost of transferring a threshold across

Variant (CAD6, 48 cases)	AUROC	Found (%)	False alarm (%)	Objs / defects / negatives	6 / 22 / 31
Full method	0.71 [0.62, 0.83]	92 [83, 100]	78 [56, 100]	AUROC (defect vs. undamaged)	0.540
No camera correction	0.90 [0.81, 0.98]	92 [83, 100]	56 [33, 67]	Detected (leave-one-object-out)	0/22
No stain model	0.73 [0.64, 0.84]	96 [88, 100]	89 [78, 100]	Located within 15 mm	0/22
Random starts	0.45 [0.37, 0.56]	38 [29, 46]	44 [22, 67]	False alarms	1/31
Fixed grid of starts	0.50 [0.41, 0.62]	33 [17, 54]	44 [11, 78]	Table 10. Real objects (SiM3D) scanned before and after damage; truth from the measured scans.	
Whole image	0.56 [0.44, 0.71]	75 [58, 92]	61 [50, 67]		

Table 8. Ablations on CAD6 (frozen settings, fixed 48-case subset).

Method	AUROC	Found (%)	SiM3D (defect vs. undamaged)	False alarm (%)	Ablated hidden (%)
v1 (frozen)	0.79 [0.68, 0.88]	54 [40, 69]	46 [33, 60]	55 [41, 69]	88 [81, 95]
v1.1 (camera-check + correction fix)	0.88 [0.79, 0.94]	66 [48, 81]	55 [41, 69]	55 [41, 69]	88 [81, 95]

Table 9. Second sealed test: 30 further Rotor37 blades (67–96), 240 cases, unused by any earlier experiment. Each method ran once with its frozen threshold.

blade designs.

A pre-registered negative result (v2). Targeting the remaining false alarms, we pre-registered three changes and a rule on development blades: a smooth surface appearance field shared by both hypotheses (A), a dent-size prior (D), or both (E), adopted only if recall at matched false-alarm rate improved by at least five points. None did: D matched v1 (67% recall, AUROC 0.87 vs. 0.85), while A and E raised negative fit gains so much that recall fell to 6%; the extra appearance freedom helped the dent hypothesis as much as the null. The change that does reduce false alarms is v1.1’s reference correction of silhouette and shading (above), which also lowers the worst development false-alarm gains on Rotor37 (2.46 to 0.09).

Physical validation (pending real validation).

Pending real validation. A 2× print of CAD6 blade 5 will be dented with steel balls (target depths 0.3, 0.7 and 1.2 mm), photographed with marker-based poses before and after denting, and measured by structured-light scan (or a 0.01 mm dial gauge). This section is reserved for those results; the rest of the paper does not depend on it.

Qualitative results and failures. [TODO: Figure: observed images, H_0 residual, recovered vs. true dent for successes, a colour/dent ambiguity case and an unobservable dent.]

Real objects. No public dataset pairs real blade photographs with measured damage geometry, so we use SiM3D (61) six wooden stools scanned with an industrial structured-light sensor before and after damage, with calibrated photographs and expert defect annotations. The healthy scan replaces the CAD model, the healthy photograph from the same scanner position serves as the reference after 2D alignment, and the measured scan difference is the truth; annotated regions with no measured shape change (paint, contamination) count as negatives. On these textured objects our method does not separate physical defects from undamaged areas (Table 10). The residual between sessions is dominated by wood-grain misregistration and specular changes that the shader, trained on 12 photographs per object, does not model. We report this as a failure case: the evidence test is only as reliable as the appearance model behind it.

6. Limitations and Conclusion

Our positive results use rendered images of public compressor geometry with synthetic damage. On real scanned objects (SiM3D) the method did not separate physical defects from undamaged areas; real-blade validation with measured damage remains future work, and an appearance model that holds on real materials is its main prerequisite. The dent model has one dent per blade, and edge damage such as nicks and material loss is out of scope. The conformal guarantee holds for negatives exchangeable with the calibration set; a new camera rig or lighting setup requires recalibration. The method answers whether these views support a physical change; it does not certify airworthiness or replace an inspector.

[TODO: conclusion paragraph after results]

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